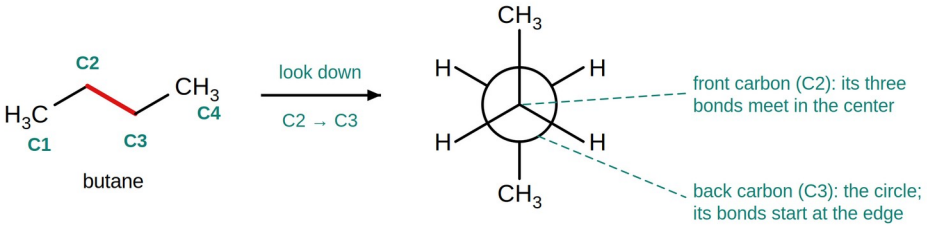
Newman Projections

Name _____ Date _____

A Newman projection is what you see when you look straight down one C–C bond. Part 1 walks you through the Newman Navigator (chemustad.org/newman.html) with butane, one step at a time. Part 2 is practice: draw each answer on paper first, then build the molecule in the Navigator and check yourself.

Reading a Newman projection

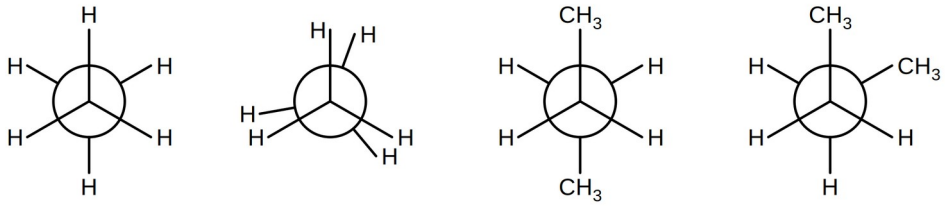


anti: the two CH_3 groups are 180° apart

Front carbon: The carbon closest to you. Its three bonds meet in the center of the drawing.

Back carbon: The carbon behind it, drawn as the circle. Its bonds start at the edge of the circle.

Dihedral angle: The angle between a bond on the front carbon and a bond on the back carbon, as you look down the C–C bond.



Staggered vs. eclipsed: In a staggered conformation the back bonds sit between the front bonds. In an eclipsed conformation they line up behind them. I draw eclipsed bonds slightly offset so you can still see the back ones. Eclipsed conformations are the energy maxima as the bond turns.

Anti vs. gauche: These describe two big groups in a staggered conformation. Anti means 180° apart, gauche means 60° apart. Gauche costs some energy because the two groups are close enough to bump into each other.

Every interaction has an energy cost. Add them up to get the energy of a conformation. These are the numbers the Navigator uses, so your answers should match what it shows.

Interaction	Cost (kJ/mol)
H/H eclipsed	4.0
CH_3/H eclipsed	6.0
CH_3/CH_3 eclipsed	11.0
CH_3/CH_3 gauche	3.8

Example: *Eclipsed ethane has three H/H eclipsed pairs, so it costs 3 × 4.0 = 12.0 kJ/mol more than staggered ethane.*

Part 1. Walk through the Navigator with butane

Open chemustad.org/newman.html on a laptop, tablet, or phone. Do each step in the Navigator, then answer on this sheet. The numbered cards (on the left on a computer, below the 3D model on a phone) match the steps here.

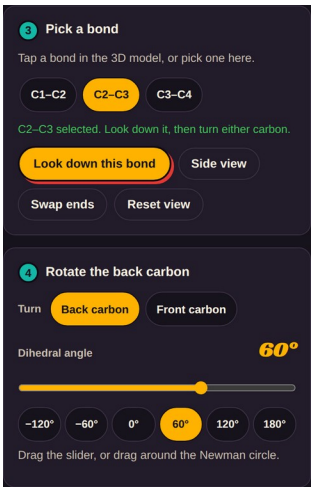
Step 1. Build the molecule. At the top of the left panel, pick Type. Under 1 Choose a molecule, enter butane and press Build. You can also find it under Presets, or draw it yourself under Draw.

(a) Draw the bond-line structure of butane and number the carbons C1 to C4.

(b) Drag the 3D model around. How many hydrogens are on C2? _____

Step 2. Find the main chain (optional). Press Find the main chain, tap the carbons from one end to the other, press Check chain, then tap the end that should be C1. This is where the numbers in a bond name like C2–C3 come from. If you already know it, press Skip, I know this.

Step 3. Look down the C2–C3 bond. Under 3 Pick a bond, tap C2–C3, then press Look down this bond.



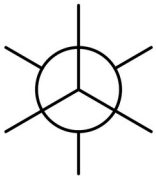
(a) Which carbon is in front? _____ Which carbon is the circle? _____

(b) Press Swap ends and read the caption under the small Newman panel. What changed, and what stayed the same? Press Swap ends again to go back.

_____

_____

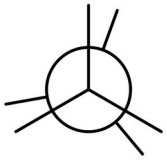
Step 4. Rotate the back carbon. Under 4, make sure Turn is set to Back carbon. Press 180°, 120°, 60°, and 0° in order. For each one, draw the Newman projection, then copy the name and energy from the result card (top right on a computer, below the 3D model on a phone).



180°

Name: _____

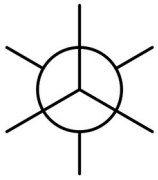
Energy: _____ kJ/mol



120°

Name: _____

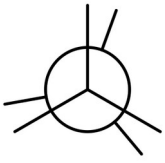
Energy: _____ kJ/mol



60°

Name: _____

Energy: _____ kJ/mol



0°

Name: _____

Energy: _____ kJ/mol

(a) Which conformation is lowest in energy? _____ Highest? _____

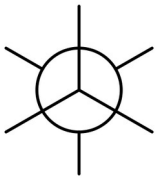
(b) Use the energy table on page 1 to show where the 19.0 kJ/mol at 0° comes from. Do the same for 120°.

_____

_____

Step 5. Turn the front carbon instead. Switch Turn to Front carbon and press 60° again. This time the back carbon stays put and the front carbon turns.

(a) Draw the Newman projection shown in the small Newman panel.



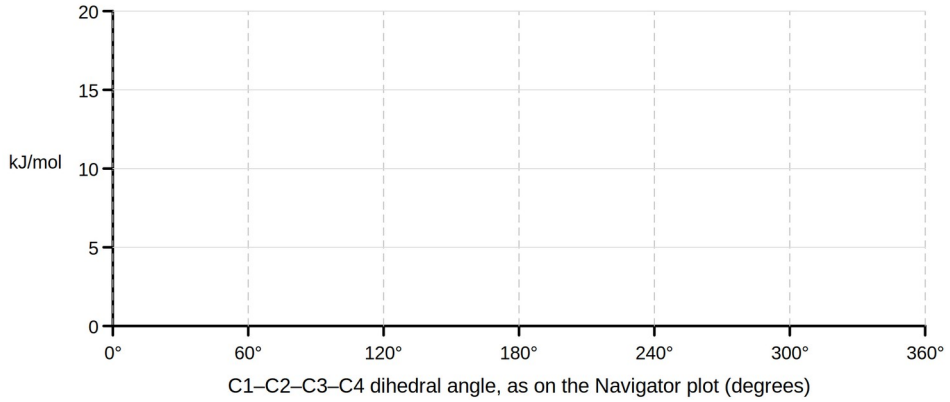
(b) Compare it with your 60° drawing from Step 4. Which groups moved? Is it the same conformation, with the same energy? Why?

_____

_____

Step 6. Read the energy plot. Scroll to 5 Torsional energy, then drag the slider slowly and watch the dot move along the curve. The plot runs from 0° to 360°, so −60° on the slider is 300° on the plot. Angles between the named ones show up as skew.

(a) Sketch the curve on the axes. Label anti, gauche, eclipsed, and totally eclipsed.



(b) Circle the points on your sketch that are transition states between conformations.

(c) How much energy does it take to go from anti to totally eclipsed? _____ kJ/mol

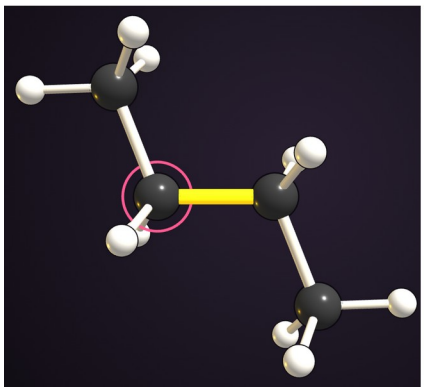
(d) Gauche butane is 3.8 kJ/mol above anti even though nothing is eclipsed. Why?

_____

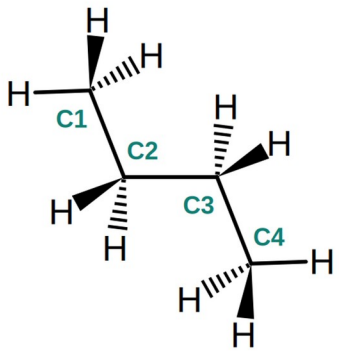
_____

Step 7. Side view. Turn the back carbon to 180° (anti), then press Side view so the C2–C3 bond lies flat across the screen. Compare the model with the drawing: bonds in the plane of the page are plain lines, wedges come toward you, and dashes go away. Each CH_2 has one H on a wedge and one on a dash.

Navigator, Side view



On paper



(a) Now press 60° and then Side view again. Draw this gauche conformation the same way, with every H shown.

(b) Which C–C bond is no longer in the plane of the page? _____

Step 8. Try it. Scroll to 6 Try it. Answer each question in the Navigator, press Check, then Next for the next one. Write down one you missed the first time and what you changed.

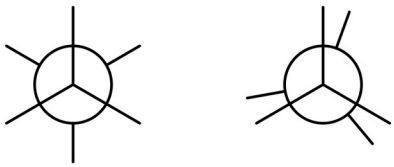
_____

_____

Part 2. Practice

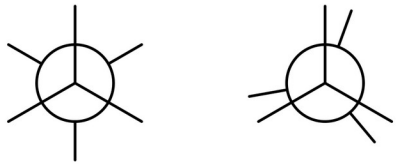
Work each problem on paper first. Then type the molecule in bold into the Navigator, look down the bond given, and check your drawing, angle, and energy.

1. Ethane. Draw the staggered and eclipsed conformations of ethane looking down the C–C bond. What is the barrier to rotation? Check with **ethane**.



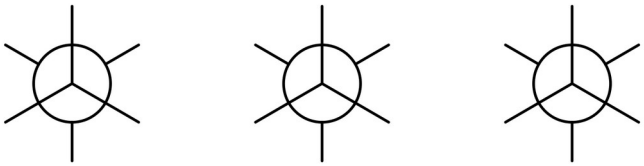
Barrier: _____ kJ/mol

2. Propane. Draw the lowest- and highest-energy conformations of propane looking down C1–C2. Add up the interactions to find the barrier. Check with **propane**, bond C1–C2.



Barrier: _____ kJ/mol

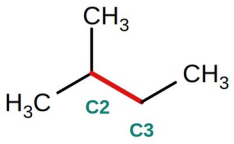
3. 1,2-Dichloroethane. Draw the anti conformation and both gauche conformations looking down C1–C2. Check with **1,2-dichloroethane**.



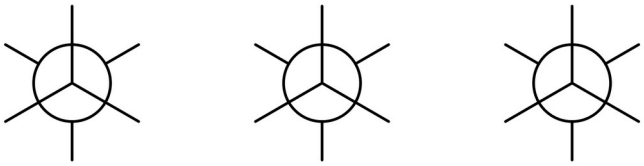
(a) Which one is lowest in energy? _____

(b) A Cl/Cl eclipsed pair costs 16.0 kJ/mol. What does the totally eclipsed conformation cost? _____ kJ/mol

4. 2-Methylbutane. Draw the three staggered conformations looking down C2–C3. Check with **2-methylbutane**, bond C2–C3. Drag the slider slowly through the full range and read the name and energy. Skip the angle buttons here: the two CH_3 groups on C2 tie, so the angle shown jumps to whichever pair is closest.



2-methylbutane

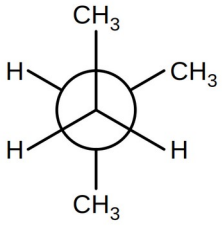


(a) Count the CH_3/CH_3 gauche interactions in each one.

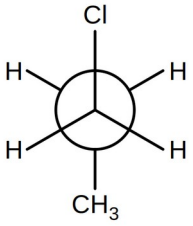
(b) Which two have the same energy? _____

(c) What is the energy difference between the lowest and highest staggered conformation? _____ kJ/mol

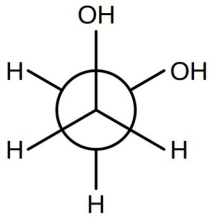
5. From Newman to bond-line. Each drawing below looks down one C–C bond of a different molecule. Draw the bond-line structure and name it. Then type the name you came up with into the Navigator and see if you can get the same view.



(a)

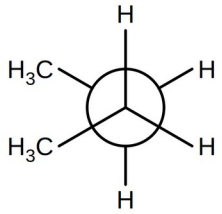


(b)

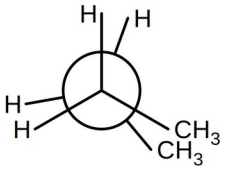


(c)

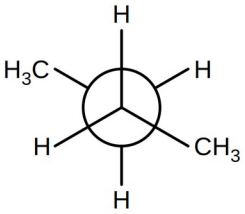
6. Rank them. All four drawings are butane looking down C2–C3, but they are not all turned the same way. Find the two CH_3 groups first. Label each one (anti, gauche, eclipsed, or totally eclipsed), then rank them from highest to lowest energy.



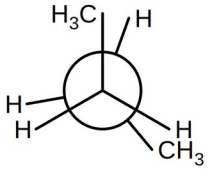
A



B



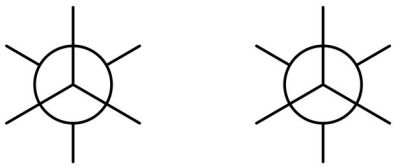
C



D

Highest _____ > _____ > _____ > _____ Lowest Which are transition states? _____

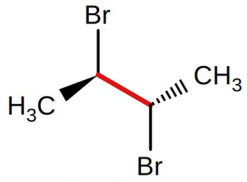
7. Ethylene glycol. Draw the anti and gauche conformations of $HOCH_2CH_2OH$ looking down C1–C2. Check with **ethylene glycol**. In the Navigator, which one is lower in energy? Why is this different from butane?



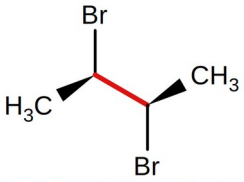
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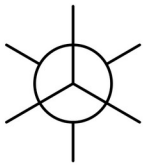
8. 2,3-Dibromobutane. For each isomer, draw the Newman projection looking down C2–C3 with the two Br atoms anti. Are the CH_3 groups anti or gauche? Check with **meso-2,3-dibromobutane** and **(2R,3R)-2,3-dibromobutane**.

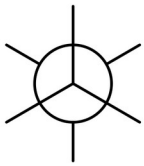


meso-2,3-dibromobutane



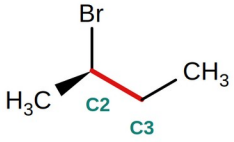
(2R,3R)-2,3-dibromobutane



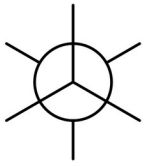


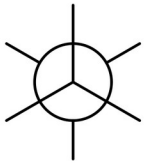
meso: CH_3 groups _____ (2R,3R): CH_3 groups _____

9. Setting up E2. An E2 reaction needs an H and the leaving group anti to each other. Looking down C2–C3 of (R)-2-bromobutane, draw the two conformations that put an H on C3 anti to Br. Which one gives trans-2-butene and which gives cis-2-butene? Which is lower in energy? Check with **2-bromobutane**.



(R)-2-bromobutane





_____

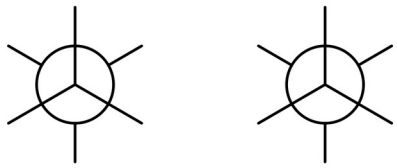
_____

10. Methylcyclohexane (challenge). Type **methylcyclohexane** and look down C1–C2.

(a) Is the CH_3 axial or equatorial? Press Flip the chair. Now?

(b) How much higher in energy is the chair with the CH_3 axial? _____ kJ/mol

(c) Draw the Newman projection down C1–C2 for each chair. Label the ring carbons C6 (on C1) and C3 (on C2). Is the CH_3 anti or gauche to C3?



(d) An axial CH_3 is gauche to two ring carbons (C3 and C5). Use the CH_3/CH_3 gauche cost to estimate how much higher in energy the axial chair is than the equatorial chair. How close is it to the Navigator's number?

_____

_____